Discrete-Event Simulation

Simulation Without a Black Box: Modeling and Simulation in the AI Era

Credits: 3 Meeting: [Days/Time, Room] Instructor: [Name, Email] Office Hours: [Times and Location]

Course Description

This course develops the theory and practice of discrete-event simulation (DES) using Python and SimPy. Students learn to build, analyze, and communicate simulation studies for complex stochastic systems. Topics include conceptual modeling, random variate generation, queueing and inventory models, output analysis, experimental design, verification and validation, and reinforcement learning for simulation-based control.

A central theme is *transparency*: every model, analysis, and visualization is expressed in readable, testable Python code — no black-box GUIs. Students are encouraged to use AI assistants as co-pilots throughout the course.

1 Textbook and Resources

Primary: *Simulation Without a Black Box: Discrete-Event Modeling and Analysis for the AI Era.* [Author]. Available as PDF on the course website.

Supplemental: Law (2015), *Simulation Modeling and Analysis*, 5th ed. (on reserve).

2 Prerequisites

- Probability and statistics (random variables, confidence intervals)
- Python programming (functions, loops, importing libraries)
- Recommended: linear algebra (for the reinforcement learning module)

3 Course Tools

Tool	Purpose
Python 3.11 + `simdes` package	Core programming environment
JupyterLab	Lab notebooks and homework submissions
Docker / conda	Reproducible environment (install instructions on course site)
GitHub Classroom	Assignment submission and peer review

4 Grading

Component	Weight
Homework (7 assignments, HW-01–07)	35%
Midterm project	20%
Final project	30%
Lab participation	10%
Peer review	5%
Total	**100%**

Individual homework weights vary (5–10%); see each assignment for details. Full grading criteria: `rubrics/`.

5 Course Policies

Academic Integrity

You may use AI assistants (ChatGPT, Claude, GitHub Copilot, etc.) to help with code, debugging, and writing. You must understand everything you submit. Submitting AI-generated work you cannot explain is an academic integrity violation.

Collaboration is encouraged on conceptual questions. **Code and written analysis must be your own.** Sharing code files or GitHub repositories before the deadline is a violation.

Late Policy

Timing	Deduction
On time	None
Up to 48 hours late	-10%
2–7 days late	-25%
>7 days late	0 (submit anyway for feedback)

No submissions accepted after solutions are posted (typically 1 week after the due date). Exceptions for documented emergencies only — contact the instructor *before* the deadline.

Accessibility

Students requiring accommodations should contact the instructor during the first week of class with their accommodation letter.

Communication

Questions about course material: post to the class discussion forum so all students benefit from the answer. Personal or grade-related questions: email the instructor directly. Expect a response within 48 hours on business days.

6 Course Schedule

See `syllabus/schedule.pdf` for the full 15-week lecture plan.

Phase	Topics
Weeks 1–4	Foundations: conceptual modeling, event-driven simulation, manual tracing
Weeks 5–11	Core DES: SimPy queues, inventory, advanced models
Weeks 12–14	Analysis: input modeling, output analysis, experimental design
Week 15	Advanced topics: V&V, simulation optimization, reinforcement learning

Key dates:

- Midterm project out: End of Week 12 **Due: End of Week 14**
- Final project out: End of Week 15 **Due: Finals week**